PROJECT REPORT

SMART HOME AUTOMATION USING ESP32 AND JAVA

Submitted in Partial Fulfillment of the Requirements for the Internship Project

Submitted By

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CERTIFICATE

This is to certify that the project entitled **"Smart Home Automation Using ESP32 and Java"** has been carried out by **Mohanraj M** of the Department of Electronics and Communication Engineering, Dhanalakshmi College of Engineering, as a part of the internship project during the academic year 2025–2026.

The project has been completed successfully under our guidance and fulfills the requirements of the internship program.

Project Guide

(Signature)

Head of Department

(Signature)

ACKNOWLEDGEMENT

I express my sincere gratitude to my internship organization for providing me with an opportunity to complete this project. I thank my project guide for valuable suggestions and continuous encouragement throughout the project.

I also thank the Head of the Department, faculty members, friends, and my parents for their constant support and motivation in successfully completing this project.

ABSTRACT

Smart Home Automation is one of the rapidly growing applications of the Internet of Things (IoT). The objective of this project is to develop a simple and cost-effective system that allows users to control home appliances remotely through a Java desktop application.

The proposed system uses an ESP32 microcontroller connected to Wi-Fi. The Java application communicates with the ESP32 using HTTP requests. Whenever the user presses the ON or OFF button in the application, a request is sent to the ESP32, which controls the connected electrical device.

The project demonstrates reliable communication between software and hardware using wireless networking. It provides an easy-to-use interface, low implementation cost, and improved convenience for users.

TABLE OF CONTENTS

CHAPTER 1 – INTRODUCTION

Home automation has become an important application of IoT technology. It enables users to monitor and control electrical appliances remotely using computers or smartphones. Traditional manual switches require physical interaction, whereas IoT-based automation allows remote operation through a network.

This project develops a Smart Home Automation system using an ESP32 microcontroller and Java programming language. The Java application sends HTTP commands over Wi-Fi to the ESP32, which receives the request and switches appliances ON or OFF.

The system is simple, efficient, inexpensive, and suitable for homes, offices, and laboratories.

CHAPTER 2 – PROBLEM STATEMENT

In conventional homes, electrical appliances are controlled manually using wall switches. This method becomes inconvenient when users are away from home or when elderly and physically challenged individuals need to operate appliances.

The absence of remote monitoring also leads to unnecessary power consumption because appliances may remain switched ON accidentally.

Therefore, an affordable and efficient automation system is required to provide wireless control over electrical devices.

CHAPTER 3 – OBJECTIVES

The major objectives of this project are:

- To design a smart home automation system.
- To establish communication between Java and ESP32.
- To control appliances using Wi-Fi.
- To reduce human effort through automation.
- To provide a simple graphical user interface.
- To improve energy efficiency.

CHAPTER 4 – EXISTING SYSTEM

The traditional system uses manual switches for controlling electrical appliances.

Disadvantages

- • No remote access.
- • Time-consuming operation.
- • Increased electricity wastage.
- • No automation.
- • Limited user convenience.

CHAPTER 5 – PROPOSED SYSTEM

The proposed system consists of a Java desktop application and an ESP32 microcontroller connected through a Wi-Fi network.

The user operates the Java application by clicking ON or OFF buttons. The application sends an HTTP request to the ESP32. The ESP32 receives the request and controls the connected relay or LED accordingly.

Advantages

- • Wireless communication.
- • Low-cost implementation.
- • Easy installation.
- • Fast response.
- • User-friendly interface.
- • Reliable operation.

CHAPTER 6 – HARDWARE AND SOFTWARE REQUIREMENTS

Hardware

- • ESP32 Development Board
- • Wi-Fi Router
- • Laptop/Computer
- • USB Cable
- • LED/Relay Module
- • Breadboard and Connecting Wires

Software

- • Java JDK 17
- • Eclipse IDE / IntelliJ IDEA
- • Arduino IDE

- Windows 10/11
- ESP32 Board Package

CHAPTER 7 – SYSTEM ARCHITECTURE

```
        +--------------------+
        |   Java Application |
        +---------+----------+
                  |
             HTTP Request
                  |
            Wi-Fi Network
                  |
        +---------+----------+
        |       ESP32        |
        +---------+----------+
                  |
             Relay / LED
                  |
          Home Electrical Device
```

Working

1. User opens Java application.
2. User clicks ON or OFF button.
3. Java sends HTTP request.
4. ESP32 receives request.
5. ESP32 switches appliance.
6. Status is displayed successfully.

CHAPTER 8 – IMPLEMENTATION

The implementation consists of two parts.

Java Application

- Developed using Java Swing.
- Provides ON and OFF buttons.
- Sends HTTP requests to ESP32.

ESP32 Program

- Connects to Wi-Fi.
- Starts HTTP server.
- Receives commands.
- Controls GPIO pins connected to relay or LED.

Communication between Java and ESP32 is achieved through HTTP GET requests over the local Wi-Fi network.

CHAPTER 9 – RESULTS

The developed system successfully controls electrical devices through the Java application.

Observations

- Wi-Fi connection established successfully.
- Java application communicates with ESP32.
- Devices respond instantly.
- ON/OFF operations execute correctly.
- System performs reliably with minimum delay.

The project demonstrates successful integration of Java programming with embedded IoT hardware.

CHAPTER 10 – CONCLUSION

The Smart Home Automation system using ESP32 and Java has been successfully developed and tested. The project enables wireless control of electrical appliances through a simple Java desktop application.

The system is reliable, economical, user-friendly, and suitable for modern smart homes. It also provides a strong foundation for developing advanced IoT-based automation systems.

FUTURE SCOPE

- Android application support.
- Voice control using Google Assistant.
- Alexa integration.
- Cloud database connectivity.
- Real-time monitoring.
- Sensor-based automatic control.
- Mobile notifications.

• Energy consumption analysis.

REFERENCES

1. ESP32 Technical Documentation.
2. Java SE Documentation.
3. Arduino IDE User Guide.
4. IoT Research Papers.
5. HTTP Communication Documentation.

End of Report